

Aayush Agarwal

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Profile Summary

Software Developer in training (B.Tech CSE, Class of 2027) with strong object-oriented design and DSA skills (400+ LeetCode problems). Experienced in iterative, test-driven development; built full-stack AI systems using clean code principles. Thrives in collaborative, feedback-driven environments — demonstrated through E-Cell IIT Bombay and cross-functional hackathon teams. Passionate about continuous learning and delivering meaningful solutions to real business problems.

Education

Bachelor of Technology in Computer Science and Engineering 2023 – 2027
I.K. Gujral Punjab Technical University CGPA: **9.43** — **Rank 1** in CS & AI Branch

Projects

Enterprise RAG AI Assistant

- Iteratively built a RAG-based assistant over 1,000+ PDF pages; applied continuous delivery practices to ship incremental improvements to retrieval accuracy and UX
- Indexed 50,000+ vector embeddings using FAISS, reducing retrieval latency by 68% (4.8s → 1.5s) through performance-driven refactoring
- Designed modular document ingestion pipeline (10+ file formats) with clean separation of concerns, achieving 89% answer relevance
- Implemented conversational memory module using OO principles, improving multi-turn query accuracy by 35% over baseline
- **Tech Stack:** Python, LangChain, FAISS, ChromaDB, Gemini/OpenAI API, Streamlit, Vector Databases

Sentiment Analysis with Fine-Tuned DistilBERT

- Applied iterative model development — baseline TF-IDF + Logistic Regression, then fine-tuned DistilBERT, achieving 91% F1-score (15% improvement)
- Built end-to-end training pipeline with PyTorch using clean, modular code; validated each iteration with confusion matrix and classification reports
- Published model on Hugging Face Hub for reproducibility and public inference, demonstrating commitment to knowledge sharing
- **Tech Stack:** Python, PyTorch, Hugging Face Transformers, Scikit-learn, Pandas, NLP

AI-Powered Smart Waste Management System

- Collaborated in a cross-functional team to build an AI waste classification system; benchmarked 4 CNN architectures iteratively, selecting InceptionV3 (91.3% accuracy, 0.89 F1)
- Integrated ESP32 IoT sensors for real-time monitoring; used continuous feedback loops to trigger automated alerts at 85% bin capacity
- Delivered analytics dashboard enabling data-driven decisions, reducing simulated collection inefficiencies by 30%
- **Tech Stack:** Python, TensorFlow, Keras, InceptionV3, OpenCV, ESP32, Firebase, IoT

Skills

- **Software Engineering:** OO Design, SOLID Principles, Test-Driven Development (TDD), Pair Programming, Clean Code, Design Patterns
- **Programming Languages:** Python, Java, C++, SQL
- **Continuous Delivery & Tools:** Git, GitHub, CI/CD practices, Agile/Iterative Development, Jupyter Notebook, VS Code, Linux
- **Machine Learning & AI:** Supervised/Unsupervised Learning, Deep Learning (CNNs, Transformers), Feature Engineering, Model Evaluation, RAG, Prompt Engineering
- **Data & Databases:** NumPy, Pandas, Scikit-learn, EDA, Matplotlib/Seaborn, MySQL, SQLite, Vector Databases
- **Mathematics:** Probability, Statistics, Linear Algebra, Optimization, Calculus
- **Core CS:** Data Structures & Algorithms (400+ LeetCode), Time/Space Complexity, OOP, DBMS

Achievements & Leadership

- Campus Executive, **E-Cell IIT Bombay** — led cross-functional teams to organize technical events and workshops, embodying Thoughtworks' value of flat, collaborative environments
- **Top 700 across 11 countries** in the Ericsson Edge Academia Program — demonstrated resilience and adaptability in a globally diverse, competitive cohort
- Member, **Training Advisory Council (TAC)** — collaborated with faculty and student teams on curriculum and knowledge-sharing initiatives
- **Ranked 1st** in Robotics and Artificial Intelligence branch; **District Topper** in a National-Level Examination
- Participated in **multiple hackathons** — strengthened rapid prototyping, agile problem-solving, and teamwork under ambiguous, time-constrained conditions
- Maintained **400+ DSA problems** solved on LeetCode with active GitHub contribution streak

Certifications

- **Introduction to Natural Language Processing** – Infosys [↗](#)
- **LLMs on Google Cloud: Retrieval-Augmented Generation on Vertex AI** – Infosys [↗](#)
- **Building AI-Powered Search with MongoDB Vector Search** – MongoDB [↗](#)
- **Introduction to Modern AI** – Cisco [↗](#)